Coding & Solution

Date	15 July 2026
Team / Submission ID	
Project Name	Emotion Detection & Learning Support Engine
Submitted By	Venu Gopal (Sole Team Member / Team Lead)
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented code and the overall solution, matching it against the functional requirements defined earlier.

Solution Summary

Field	Details
Repository Link / URL	[Add GitHub repository link here before submission]
Programming Language(s)	Python 3
Framework(s) Used	Streamlit (UI), TensorFlow/Keras (BiLSTM), PyTorch + HuggingFace Transformers (BERT), NLTK, pandas, Plotly, google-generativeai
Key Features Implemented	Field & problem input with quick examples; BiLSTM and BERT emotion prediction with confidence scores; mixed-emotion detection and keyword boosting; Gemini-generated personalised response with regenerate option; rule-based fallback response; CSV interaction logging; Plotly analytics dashboard (emotion distribution, emotional journey, field-wise breakdown, summary metrics); session history clearing.
Pending / Incomplete Features	No automated test suite; no persistent database (CSV only); no user authentication / multi-user history; no README / setup guide in the repository; CSV filename used by the sidebar counter (emotion_responses.csv) does not match the file the logger actually writes to (emotion_response_examples.csv).
Setup / Run Instructions	1) Create a virtual environment and run: pip install -r requirements.txt 2) Create a .env file with GEMINI_API_KEY=your_key_here 3) Run: streamlit run app.py 4) Open the local URL Streamlit prints (default http://localhost:8501)

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Partial
4	Error handling is implemented for critical operations	Yes (broad in places)
5	The application runs without critical errors	Yes, in local testing
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

The prediction and AI-response logic is well separated from the UI, which makes the codebase easy to extend (e.g. swapping Streamlit for a REST API). The main improvement areas are documentation (a README is missing) and tightening a couple of small inconsistencies noted above.